Research Design & Final Presentations

Week 16 · Practice module · Textbook Chapter 15

Brian C. Keegan, Ph.D.
Associate Professor, Department of Information Science
University of Colorado Boulder

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This week at a glance

We close the course by turning **technique into knowledge**: how to design a research project around web data — and then present your own.

Monday | Concepts

- From question to data source; matching designs; sampling, bias & validity

Wednesday | Project workshop

- Match *your* source to a design, scope feasibility, and trade peer feedback — no new code

Friday | Final presentations

- Present your project — the last day of class

Companion notebook

`ch-15-research-design.ipynb`

Last week of the course

Friday (Dec. 4) is the final class — campus runs a Monday schedule. The **project repo & write-up are due Fri. Dec. 11.**

1. Monday · Concepts

Technical skill is not enough

You now command the whole toolkit: scrape static (Ch. 6) and dynamic (Ch. 8) pages, parse PDFs (Ch. 9), query APIs from Wikipedia to Spotify to the Fed (Ch. 10–12), apply NLP and LLMs (Ch. 13), and automate collection (Ch. 14).

But **technical skill without research design produces data, not knowledge.** A flawless scraper pointed at the wrong question still answers nothing.

This week steps back from code to the questions that decide whether excellent scraping yields real insight.

Four questions design must answer

- What is the **right question** for this source?
- What can this data **actually** tell you?
- What are the **threats to validity**?
- How do you **communicate** findings responsibly?

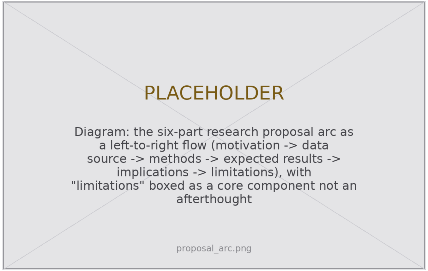
Textbook Ch. 15, "Technical Skill Is Not Enough." The tools are means; knowledge is the end.

From question to data source: the proposal arc

The best projects start with a **question, not a dataset** — the source serves the question. Every proposal (class project, thesis, paper, grant) shares one arc:

1. **Motivation** — what problem, why it matters, who benefits
2. **Data source** — which web data, what constructs, what gaps
3. **Methods** — how you collect & analyze it
4. **Expected results** — what analyses & figures
5. **Implications** — what stakeholders should do
6. **Limitations** — what the data *cannot* tell you

Limitations are not an afterthought — naming what your data cannot do is as important as showing what it can.



The same arc structures a proposal, a write-up, and a talk.

Textbook Ch. 15, "The Proposal Arc."

Matching questions to designs

Your question implies a design; the design implies which chapters' tools you reach for:

Design	Web data methods	Example question
Longitudinal tracking	Wayback Machine (Ch. 7), Actions (Ch. 14)	How has a privacy policy changed over time?
Cross-sectional snapshot	API queries (Ch. 10–12)	How does Reddit moderation differ across subreddits?
Content analysis	Scraping (Ch. 6) + LLM coding (Ch. 13)	What topics dominate Colorado legislation?
Network analysis	Link & social-graph extraction (Ch. 10, 12)	How connected are climate articles on Wikipedia?
Natural experiment	Event-triggered collection (Ch. 14)	How did a policy change shift search patterns?

Design considerations vary. Longitudinal: Wayback snapshots are uneven; scrapers need upkeep. Cross-sectional: **selection bias** — you get only platforms that exist and expose the variable. Content analysis: validate automated coding against a hand-coded sample. Natural experiment: causal claims are always tentative — you need a clear treatment and a defensible comparison group.

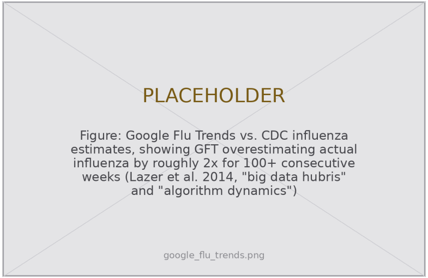
Textbook Ch. 15, "Matching Questions to Methods" & "Common Research Designs with Web Data."

Web data is not a random sample

Every design inherits one problem: web data is a **convenience sample**, shaped by who uses the platform, what the API exposes, and what your scraper can reach.

- **Who is missing?** Reddit skews young, male, American; Wikipedia editors skew educated & Global North.
- **What is hidden?** APIs expose what platforms *choose* to — not deleted posts, shadow-bans, or ranking algorithms.
- **What is lost?** Pages are deleted and redesigned; yesterday's data may already be gone (Ch. 3).

Construct validity: does the thing you can measure (pageviews, posts, queries) match the concept you claim to study (attention, opinion, illness)? Weak construct validity is the most common flaw.



Google Flu Trends overestimated flu $\sim 2\times$ for 100+ weeks: *big-data hubris* plus *algorithm dynamics* — the instrument drifted while the phenomenon held still.

Textbook Ch. 15, "Sampling and Representativeness." Construct validity: Salganik (2018); the cautionary tale: Lazer et al. (2014).

Human subjects & the IRB

If your data is about **people** — and most web data is — you may be doing **human subjects research.** That is not a call you make for yourself.

- **Public does not mean exempt.** "It was on the public web" is about access, not ethics. The **IRB** makes the determination — if you think "obviously fine, I won't ask," that is exactly when to ask.
- **Terms of service are not an ethics review** (Ch. 2) — a legal matter, separate from the human-subjects question.
- **Minimize & de-identify by default** — collect only the fields you need; hash or drop the rest.

Consult early when you…

- interact with users (even a message)
- collect from private / invite-only spaces
- identify specific individuals
- measure sensitive attributes (health, sexuality, immigration, politics)

This class: your instructor is the checkpoint.

Publishing (5617): request an IRB determination early.

Textbook Ch. 15, "Human Subjects and the IRB." See Fiesler et al. (2020) and the ethics appendix of Salganik (2018).

2. Wednesday · Project Workshop

Wednesday: design your own project

Today is a **workshop**, not a notebook. You proposed a project back in Week 10 — now we harden it into something you can execute, defend, and present.

Three activities, in pairs:

1. **Draft** — write your project against the six-part proposal arc.
2. **Scope** — match your source to a design and stress-test its feasibility.
3. **Review** — trade drafts and give kind, specific, actionable feedback.

Bring what you have: a question, a candidate data source, and any data you have already pulled this semester.

The goal

Leave today with a design you could **write down in full** — because a design you cannot write down is a design you have not finished.

Textbook Ch. 15, "Managing the Research Process." Feedback ethos reused from the Friday revision workflow.

Activity 1 — draft your six-part design

Fill in one crisp sentence for each. Vague answers now become dead ends later:

1. **Motivation** — "I want to know __________ because ________."
2. **Data source** — name the site/API and the **construct** each field measures.
3. **Methods** — which chapters' tools, and in what order.
4. **Expected results** — the actual figure or table you picture.
5. **Implications** — who acts on the answer, and how.
6. **Limitations** — the honest "this cannot tell you…".

One page, six boxes — if a box is blank, that is where the project is weakest.

Textbook Ch. 15, "The Proposal Arc." Exercise 1: a 1,500-word version of exactly this.

Activity 2 — scope feasibility

First, **name your unit of analysis** — is a row a post, a user, a day, or a community? Deciding after collection means collecting again.

Then run the **pitfalls checklist** against your design:

- Am I **sampling on the outcome**? (only viral posts $\Rightarrow$ can't explain virality)
- Bot & spam **contamination** filtered?
- **Survivorship** — what deleted/suspended content is invisible?
- Will I **change instrumentation** mid-collection? (self-inflicted drift)

Can you actually collect it?

- Rate limits & volume — does it finish in time?
- ToS & `robots.txt` clear? (Ch. 2)
- IRB checkpoint needed?
- Erosion risk — capture **now** or lose it? (Ch. 3)

Validate early

Wire a `validate_dataset()` check into collection *now*, not at analysis — or you discover a June redesign emptied half your columns in September.

Textbook Ch. 15, "Common Pitfalls" & "Data Quality and Validation."

Activity 3 — peer review

Swap designs with your partner. Read it as a friendly skeptic and answer four questions in writing:

1. Is the **question actually answerable** with this source?
2. What is the **single biggest threat to validity**?
3. What is **missing, hidden, or lost** in this sample?
4. One **concrete** thing that would make it stronger.

Same ethos as our Friday code reviews: **kind, specific, actionable** — name the line and the claim, assume good faith, end with a clear verdict.

Give feedback that helps

- Point to the *design*, not the person
- Separate “must fix” from “nice to have”
- Offer a fix, not just a flaw

You get the same in return — take one reviewer’s biggest concern and revise before Friday.

Textbook Ch. 15, “Sampling and Representativeness” as the review lens.

Reproducible from day one

Your final deliverable is a **repository**, so structure it like one now — someone else (or you in six months) should re-run it and get the same results:

```
my-project/
  .gitignore # keep keys & big data out
  README.md # overview + setup
  requirements.txt # PINNED versions
  data/
    raw/ # scraped data, NEVER edited
    processed/ # cleaned data
    README.md # datasheet lives here
  notebooks/ # 01_collection ... 03_analysis
  scripts/ # scraper.py
  output/ # figures/ tables/
```

Write a datasheet

For every dataset, record:

- **What** each variable measures & units
- **How** — tools, params, filters, date range
- **Limits** — what is not captured; likely biases
- **Ethics** — PII? ToS-compliant? safe to share?

Openness is a **design choice:** pinned environments and archived raw data stay verifiable even as platforms erode.

Textbook Ch. 15, "Reproducibility" & "Data Documentation." Datasheets: Gebru et al. (2021). Git: *Missing Manual* Ch. 31.

A design you cannot write down
is a design you have not finished.

Specify the question, the source, the unit,
and the limitation — *before* you collect.

The discipline of writing it down *is* the method.

3. Friday · Final Presentations

Friday: final presentations

The last day of class — you present the project you designed this week.

Logistics

- $\sim$6 minutes per team + 2 minutes of questions
- Follow the **proposal arc**: motivation $\rightarrow$ data $\rightarrow$ methods $\rightarrow$ results $\rightarrow$ implications $\rightarrow$ limitations
- Slides or a live notebook — your choice; **practice the timing**
- Everyone presents; everyone asks at least one question

Two deadlines

Today (Dec. 4) — present your work in class.

Fri. Dec. 11 — the polished **project repository & write-up** are due. Present now; harden the write-up with the week's feedback.

Textbook Ch. 15, "Communicating Findings." The talk is the proposal arc, spoken.

What a strong final presentation looks like

Do

- Lead with the **“so what?”** — the finding, not the plumbing
- State a **claim**, then show the evidence for it
- Be **honest about limitations** — it reads as strength, not weakness
- Know your **audience** & its evidentiary bar
- Show your work: methods reproducible from the description

Same finding, three audiences

- **Academic** — precision, explicit limits, prior work; replicable methods
- **Policy brief** — actionable, plain language; lead with implications
- **Data journalism** — narrative, visual evidence, accountability framing

Pick the register that fits *your* question — and commit to it.

Textbook Ch. 15, “Communicating Findings.”

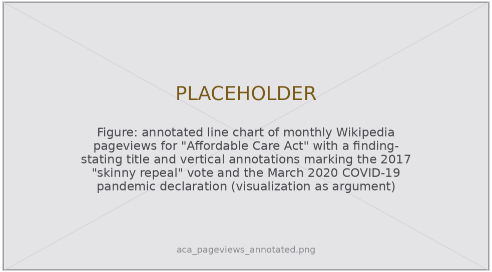
Your visualizations are arguments

A default `plt.plot()` makes the reader **guess.** A designed figure **argues:**

- Title states a **finding**, not the axes ("…spikes around legislative and health crises," not "Pageviews over time")
- **Annotate** peaks with the events that caused them
- Axis labels carry **units**
- Kill chart junk — drop the top & right spines

Test: if a reader saw *only* this image and its title, would they get your claim?

A model to steal from: ProPublica's *Dollars for Docs* — question → scrape hidden disclosures → clean & link → public database. Journalists publish code, describe cleaning, and state limits — exactly our reproducibility standard.



The title makes the claim; the annotations supply the evidence.

Textbook Ch. 15, "Visualization as Argument" & "Data Journalism as a Model."

The deliverable — and the values behind it

Due Friday, December 11 — a repository plus a write-up:

- Reproducible repo: `README`, pinned `requirements.txt`, untouched `data/raw/`, a datasheet
- Write-up structured as the six-part arc, limitations included
- Figures that argue; methods replicable from the text

The tools you learned are **means, not ends**. The end is knowledge that serves three values from Ch. 3:

Openness, Oversight, Ownership

- **Openness** — publish code, data (where ethical), methods so others can build on it
- **Oversight** — does it help citizens & journalists understand systems that affect them?
- **Ownership** — are you building reusable infrastructure, or extracting without contributing back?

Textbook Ch. 15, “Public Interest Connection” & “Social History and Public Interest.” Values from Ch. 3.

Key takeaways

1. **Start with a question, not a dataset —** the source serves the question, and the proposal arc runs motivation → limitations.
2. **Match the design to the claim —** longitudinal, cross-sectional, content analysis, network, or natural experiment, each with its own threats.
3. **Web data is a convenience sample —** ask who is missing, what is hidden, what is lost, and whether your variable has **construct validity**.
4. **"Public" is not "exempt."** Consult the IRB, minimize and de-identify, and document the dataset with a datasheet.
5. **Communicate responsibly —** reproducible pipelines, honest limitations, and figures that argue in service of openness, oversight, and ownership.

Reading: Textbook Ch. 15. Related: Salganik (2018); Gebru et al. (2021); Lazer et al. (2020); Atteveldt et al. (2022); Rogers (2019).

That is the course.

You can reach almost any data on the web —
and you know when you *should*, what it can mean,
and how to tell someone honestly.

Go make the web legible. Thank you.